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RESEARCH

Enhancing team sports performance: a case study of the optimal post-warm-up time window

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Abstract
Warm-up routines are crucial for performance enhancement in team sports, with their effectiveness varying based on individual profiles, exercise mode and timing relative to the activity. This study investigates the impact of a 5–10 min interval post-warm-up on physical performance and sympathetic stimulus, as indicated by salivary biomarkers, in 19 college athletes. Performance was assessed via countermovement jump, 20 m sprint and handgrip strength tests conducted pre-warm-up, 5 min post-warm-up and 10 min post-warm-up. Saliva samples were collected at each stage to evaluate flux, α-amylase activity and protein content. Significant effects ($p < .001$) were observed in all three performance measures and saliva markers. Notably, sprint performance showed the greatest enhancement at the 5 min mark post-warm-up, while jumping improvements persisted through the 10 min window. However, a slight decline in sprint ability was observed from 5 to 10 min post-warm-up. These findings underscore the importance of optimizing warm-up timing and eventually considering re-warm-up strategies to maximize acute performance in team sports.

Keywords Pre-match routines · Potentiation · Jumping · Running

Introduction

Warm-up (WU) is a traditional and well-accepted team sports' pre-match routine that has been proven to be effective in reducing injury risk [1] and enhancing players' readiness for competition [2]. Consequently, several mechanisms and factors have been explored to identify the most adequate strategies for enhancing acute performance in each particular context [3]. Under this scope, intensity, volume and/or exercise sequence are considered key variables that may be manipulated during WU, promoting a wide spectrum of performance changes in individuals with different training backgrounds [4].

Acute performance potentiation is strongly linked to temperature and neuromuscular mechanisms [5] and its magnitude appears to increase when the main activity is preceded by maximal or sub-maximal efforts [6]. Even though performance seems to be enhanced when short-term high-intensity

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components are included in WU [7], the understanding of the underlying mechanisms is still limited. Moreover, caution should be applied when interpreting these reports, since most of the studies are based on short-term high-intensity preconditioning activities often linked to post-activation potentiation. In fact, the role of this phenomenon in real-world WU scenarios is questionable, as the majority of the studies that show improved performance fail to confirm the presence of post-activation potentiation mechanisms [7].

Muscle temperature, increased neural drive and fiber water content are some potential factors that may enhance short-term voluntary neuromuscular performance, a phenomenon recently identified as PAPE (i.e., post-activation performance enhancement) [5]. When looking at the role of PAPE within WU activities, the potentiation time course is one of the most important topics to consider [6]. In fact, different potentiation temporal profiles may emerge from these conditioning routines, based on the type of activity, training experience and/or players' physical profiles [5]. Additionally, the recovery time after a conditioning activity plays a significant role in the performance outcome [8], which means that pre-match routines may improve, be trivial or even inhibit performance enhancements according to the time interval between the end of WU and the beginning of the match.

Establishing windows of opportunity following conditioning activities with optimal durations is essential to guarantee that potentiation effects overcome fatigue [9]. However, as previously underlined, defining standard time intervals after pre-match routines is not viable, as players' different backgrounds may result in high inter-individual responses [10]. Currently, literature focused on the role of recovery periods after conditioning activities is conflicting. Considering the nature of the activity and players' training background, short, moderate and extensive interval times, ranging from 5 to 19 min, have been reported to promote potentiation effects [6]. As a result, medium-term recovery times (i.e., 7–10 min) have been proposed as general guidelines to induce higher potentiation after moderate-intensity conditioning activities [6]. However, most of the studies leading to these recommendations are based on closed environments such as specific strength exercises performed at moderate- to high-intensity loads.

While strength routines with external loads are extensively studied for their role in inducing PAPE [5], this study aims to investigate the potential of traditional WU routines, such as low-intensity running, dynamic stretching, linear sprints and changes of direction, to induce PAPE in real-world team sports scenarios. In fact, these routines, which do not involve additional loading conditions, are also known to increase the range of motion and muscle temperature and enhance neuromuscular readiness [11]; [3]. In addition to examining the occurrence of PAPE, it is important to explore the time window of its manifestation, since little is known about the time course effect of real-world team sports' WU routines in short-term physical performance. The goal of this study was to investigate the potentiation effect time course after a traditional team sports' WU routine on strength, salivary biomarkers and jumping and running performance of college athletes.

Methods

Experimental approach to the problem

To study the optimal performance enhancement time window after WU in team sports, 19 college athletes participated in a single day design. Participants had no injuries reported during the last 3 months previous to the study. Baseline physical and biochemical values were evaluated immediately before a traditional team sports' WU. After that, two more moments were included for assessing the potentiation time course: 5–10 min after the end of the WU. Physical performance included strength, jumping and running values and biochemical markers included saliva samples.

Participants

Nineteen college athletes participated in the study (age: 19.94 ± 2.00; stature: 171.32 ± 10.58 cm; body mass: 6.787 ± 11.57 kg; ≥ 8 h a week of regular physical activity in the following team sports: basketball, football, handball, swimming and fitness activities). Participants were recruited 15 days before the beginning of the study. Athletes with existing injuries at the beginning of the study were excluded. They were free to withdraw from the study at any time without any penalty. Written informed consent was obtained from all participants. The investigation was approved by the local institutional research ethics committee and conformed to the recommendations of the 1964 Declaration of Helsinki.

Procedures

The participants performed, on a single day, a standard team sports' WU in an indoor court. Participants were tested on three consecutive moments: before WU (Pre-WU); 5 min after WU (Post-5'); and 10 min after WU (Post-10'). This time window was established based on the optimal time course effects reported in PAPE-related research. This research indicates that the effects of PAPE, as demonstrated in multiple studies, typically manifest within a period of 6–10 min [5]. Also, coaches typically report that real-world scenarios demand for the beginning of the match, usually between 5 and 10 min after the end of the WU. Physical performance was assessed by vertical

jump capacity, linear sprint ability and handgrip strength. Saliva was sampled at each moment, and flux, α-amylase activity and protein content were evaluated. The protocol timeline is presented in Fig. 1.

WU included standard activities recognized by literature and coaches as traditional team sports WU routines, such as low-intensity running, dynamic stretching, dynamic strength exercises, linear sprints, accelerations and deceleration activities and changes of direction. Particularly, the WU was structured with the following exercises: 2 min jogging at 2/4 pace, 4 × 20 m side run, dynamic stretching for main locomotive lower limb muscles (1 × 10 hip adduction; 1 × 10 hip abduction; 1 × 10 hip flexion; 1 × 10 hip extension; 2 × 10 butt kicks while jogging; 2 × 10 knee raises while jogging; 2 × 10 straight leg march), dynamic strength exercises (2 × 10 deep squats and 2 × 10 forward lunges) and incremental intermittent sprints and changes of direction (1 × 10 m 2/3pace; 1 × 20 m 2/3 pace; 1 × 20 m with 10 decelerations with both feet; 1 × 20 m with 10 decelerations with one foot; 1 × 20 m with 4 changes of direction (90°) at 2/3 pace; 1 × 10 m full pace; 1 × 20 m full pace). The structure of the WU was intentionally designed to reflect traditional routines commonly used in team sports. Each component of the WU, from low-intensity running and dynamic stretching to linear sprints and changes of direction, was chosen not only for its role in increasing body temperature, but also for its potential to enhance specific aspects of physical performance such as range of motion, neuromuscular performance and mental readiness [3].

Countermovement jump (CMJ) was used to test vertical jump capacity and jump height was calculated using an Ergotester (Globus, Codogne, Italy). Subjects were asked to place their hands on the hips and to keep feet placed shoulder width apart in the standing position. After that they were instructed to jump as high as possible with a rapid self-selected countermovement and to land close to the take-off point. Each participant performed two attempts and the best trial from each participant was recorded for data analysis.

The 20 m sprint was used to evaluate linear sprint ability, for which a photocell system was used (Witty, Microgate, Italy). The participants stood 0.5 m behind the start line and were instructed to run as fast as possible to the 20 m mark.

Handgrip strength was tested with Jamar handheld dynamometer (Alimed, Inc., Dedham, MA) for the dominant hand. The test was performed in an upright position, with the upper arm perpendicular to the trunk, the elbow without flexion and the forearm and hand in the zero position. The test was performed two times and the highest value was noted. Results were expressed as kilograms of force. Handgrip measurements are reliable, easy and fast to perform [12], which is an advantage when evaluating athletes in real-world scenarios.

Saliva biomarkers can be useful to understand physiological phenomena associated with the activation of the autonomic nervous system [13]. Being innervated sympathetically and parasympathetically, salivary glands respond to nervous stimuli changing saliva composition [14]. For instance, saliva elicited by sympathetic stimulation is typically low in volume and high in protein content, mainly due to increased exocytosis of salivary proteins from salivary cells [14]. Physical exercise is a strong activator of the sympathetic nervous system [15] and increases in salivary total protein and/or alpha-amylase activity are common features [16–18], which can also be associated with the intensity of the exercise [16, 19].

Unstimulated whole saliva was collected in three consecutive moments for each participant, by direct draining into an ice-cold collection tube (pre-weighed) for 3 min. Subjects refrained from eating and drinking for at least 1 h before collection. After saliva collection, tubes with the samples were weighted (for saliva flux evaluation, mL/min), centrifuged at 1.500*g*, for 10 min, to remove food and cell debris and the

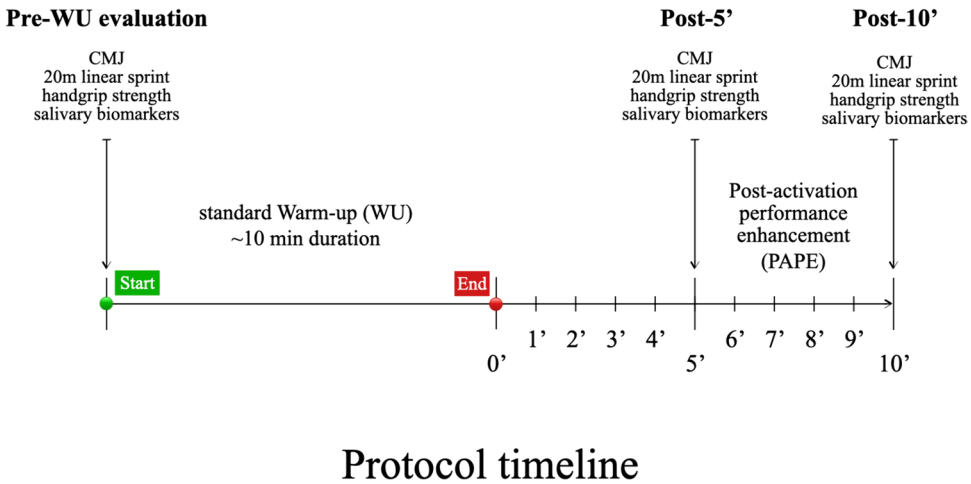

Fig. 1 Schematic representation of the experimental design. Evaluations included vertical jump capacity, linear sprint ability, handgrip strength, saliva flux, saliva *α*-amylase and saliva protein

supernatant was stored at – 20 °C until analysis. Preceding analysis, saliva samples were thawed on ice and centrifuged for 30 min, 4 °C, 13,000*g*, for removal of mucinous material [13]. Supernatant total protein concentration was assayed using the Bradford method [20].

Dinitrosalicylic acid assay was used for measuring the starch-hydrolyzing activity of salivary α-amylase [21], minimized for 96-well microplates. The reaction mixture consisted of 30 µL of 1% starch solution, 30 µL of 20 mM phosphate buffer (pH 7.0) and 15 µL of saliva (diluted to a protein concentration of 10 µg/ml in the same phosphate buffer). After incubation at 37 ºC for 20 min, the reaction was stopped by the addition of 30 µL DNS reagent (1% dinitrosalicylic acid, 0.2% phenol, 0.05% sodium sulfite, 1% sodium hydroxide). Samples were heated to 90ºC for 30 min and subsequently cooled down to room temperature. Further, 60 µL mL of sodium and potassium tartrate (40%) were added to samples. Absorbance was measured using a microplate spectrophotometer at 530 nm. A blank was prepared using the identical method, except the enzyme solution, which was added to the mixture after the addition of the DNS solution. The absorbance values were then converted to glucose equivalent using a standard curve (0.2–5.0 mg/mL). The results were expressed as mmol glucose/min/mg protein and as mmol glucose/min/L saliva.

Statistical analysis

After preliminary inspections for distribution and assumptions, a repeated measures analysis was processed to identify the effect of the time window (Pre-WU, Post-5' and Post-10') on the considered variables. Pairwise differences were assessed with Bonferroni post hoc. The statistical analysis was performed using the Statistical Package for the Social Sciences software (SPSS, Inc., Chicago, IL, USA), and statistical significance was set at $p < 0.05$.

An estimation technique approach was carried out to overcome the shortcomings associated with traditional N–P null hypothesis significance testing [22, 23]. Estimation plots were used as descriptive statistics of the pairwise differences. This graphical representation shows the individual and mean group values for the time windows and the difference of means with 95% of confidence intervals. Also, Cohen's $d_{unbiased}$ (d_{unb}) with 95% confidence intervals (CI) as effect size (ES) (an unbiased estimate has a sampling distribution whose mean equals the population parameter being estimated) was applied to identify pairwise differences [22]. Thresholds for effect size statistics were: 0.2, 0.5, and 0.8 for small, medium and large [24].

Results

The summary of findings from repeated measures for time window factor is presented in Table 1. Also, individual and differences of mean values from team-window pairwise comparisons are shown in estimation plots: Fig. 2 for physical performances and Fig. 3 for saliva measures. Complementarily, d_{unb} values with 95% confidence intervals for all pairwise comparisons are presented in Fig. 4.

The time windows factor showed significant effects in CMJ ($F = 8.2$, $p < 0.001$, $\eta^2p = 0.33$), Sprint-20 m ($F = 11.3$, $p < 0.001$, $\eta^2p = 0.39$). For these results, post hoc differences were identified for the pairwise Pre-WU vs. Post-5' and Pre-WU vs. Post-10'. There was a small increase in CMJ from Pre-WU to both Post-5' (mean difference with 95% CI = 0.03 [0.02, 0.05]; $d_{unb} = 0.41$ [0.15, 0.69]) and to Post-10' (mean differences with 95% CI = 0.03 [0.01, 0.05]; $d_{unb} = 0.37$ [0.10, 0.68]). Trivial results were identified when compared Post-5' vs. Post-10'. For 20-m sprint test, players showed a large increase in the performance from Pre-WU to Post-5' (mean difference with 95% CI = − 0.23 [− 0.34, − 0.12]; $d_{unb} = -0.99$ [− 1.59, − 0.45]) and a moderate increase from Pre-WU to Post-10' (mean difference with 95% CI = − 0.16 [− 0.27, − 0.04]; $d_{unb} = -0.60$ [− 1.09, − 0.15]). There was a small decrease in the performance test from Post-5' to Post-10' (mean difference with 95% CI = 0.07 [− 0.01, 0.16]; $d_{unb} = 0.30$ [− 0.05, 0.66]). Regarding to the handgrip performance, players presented a late small increase of ~ 2 kg from Pre-WU to Post-10'.

Table 1 Descriptive (mean ± SD) and inferential analysis (repeated measures)

Variables	Pre-WU	Post-5'	Post-10'	*F*	*p*	η^2p	Post hoc
CMJ (m)	0.35 ± 0.07	0.38 ± 0.07	0.37 ± 0.07	8.2	< .001	.33	a, b
Sprint-20 m (s)	3.83 ± 0.23	3.60 ± 0.21	3.68 ± 0.26	11.3	< .001	.39	a, b
Handgrip (kg)	42.42 ± 8.56	44.11 ± 8.9	44.5 ± 8.44	2.1	.14	.11	–
Flux (mL/min)	0.77 ± 0.38	0.34 ± 0.23	0.34 ± 0.18	30.6	< .001	.64	a, b
α-Amylase (mmol/min/L)	456.05 ± 173.41	1010.05 ± 557.99	1166.18 ± 651.18	15.6	< .001	.49	a, b
Protein (mg/mL)	1028.21 ± 477.65	2292.79 ± 1464.49	2295.81 ± 1506.99	12.2	< .001	.43	a, b

Pre-WU pre-warm-up test, *Post-5'* 5-min test, after warm-up, *Post-10'* 10-min test, after warm-up

Post hoc differences are identified as: a) Pre-WU vs. Post-5'; b) Pre-WU vs. Post-10'

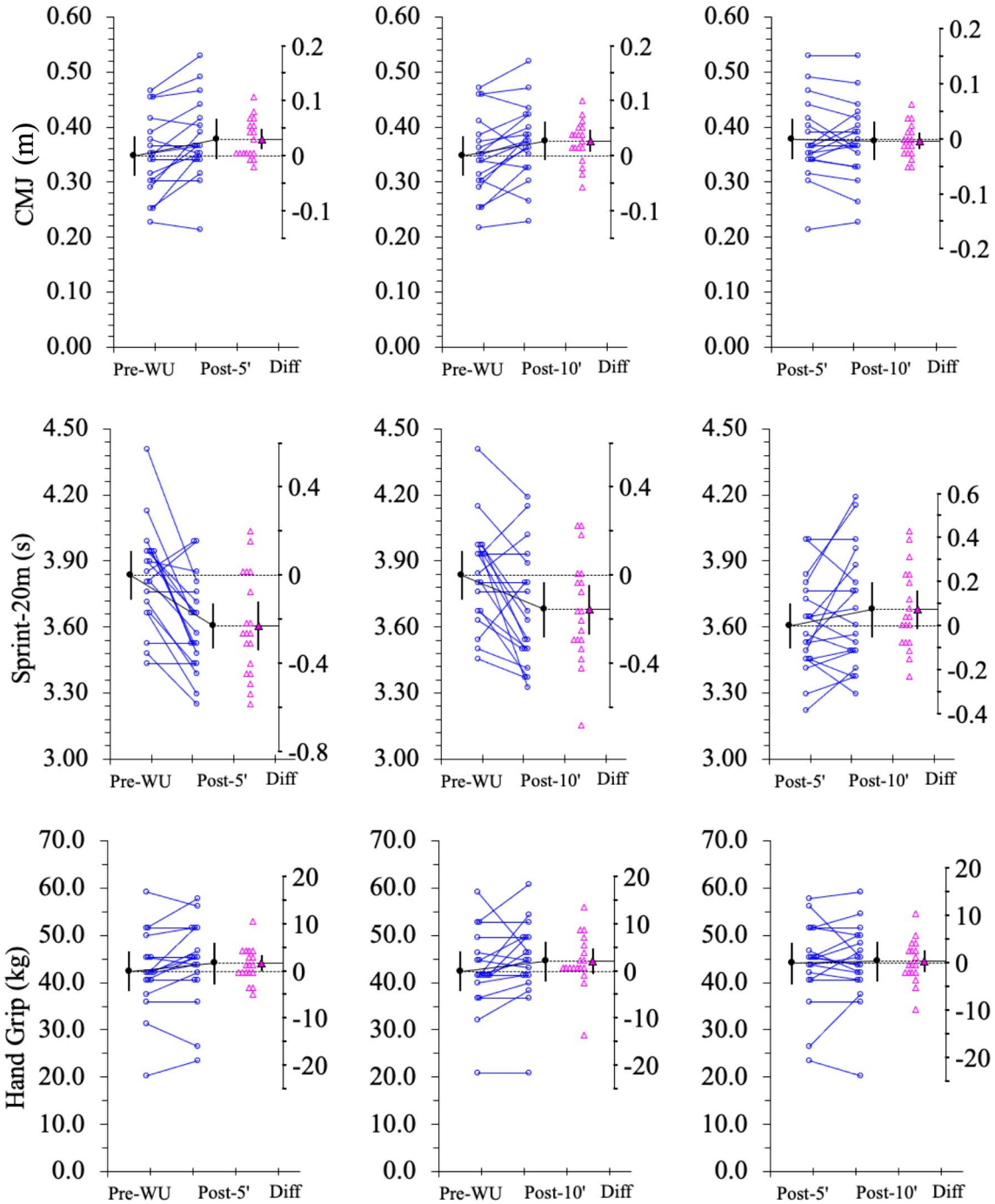

Fig. 2 Paired data showing means and 95% confidence intervals of pretest and posttest physical performance measures. The mean paired difference is shown with its 95% confidence interval against a floating difference axis, whose zero is lined up with the pretest mean. The paired data are shown as small circles joined by lines. The differences are shown as triangles on the difference axis

Considering the saliva measures, the time windows factor showed significant effects in salivary flux ($F = 30.6$, $p < 0.001$, $\eta^2p = 0.64$), α-amylase activity (mmol/min/L) ($F = 15.6$, $p < 0.001$, $\eta^2p = 0.49$) and total protein concentration (mg/mL) ($F = 12.2$, $p < 0.001$, $\eta^2p = 0.43$). Flux depicted a large decrease from Pre-WU to both Post-5' (mean difference with 95% CI = − 0.42 [− 0.57, − 0.28]; $d_{unb} = -1.28$ [− 1.95, − 0.71]) and to Post-10' (mean differences with 95% CI = − 0.43 [− 0.6, − 0.26]; $d_{unb} = -1.37$ [− 2.13, − 0.72]). For both α-amylase and protein, large increase was identified from Pre-WU to both post-5' (α-amylase: mean difference with 95% CI = 554.00

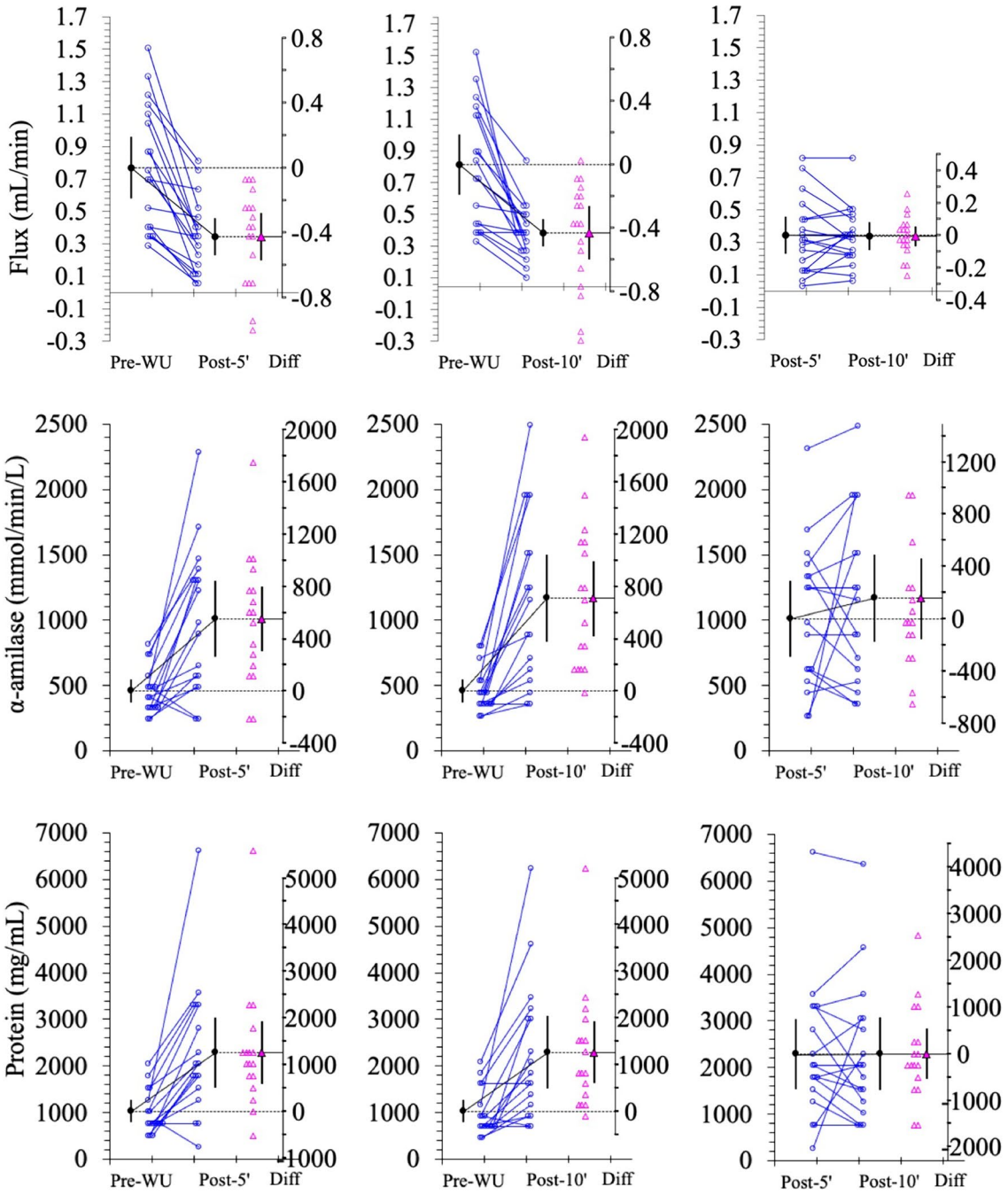

Fig. 3 Paired data showing means and 95% confidence intervals of pretest and posttest saliva measures. The mean paired difference is shown with its 95% confidence interval against a floating difference axis, whose zero is lined up with the pretest mean. The paired data are shown as small circles joined by lines. The differences are shown as triangles on the difference axis

[304.19, 803.81]; $d_{unb} = 1.28$ [0.60, 2.05]; protein: mean difference with 95% CI = 1264.58 [592.50, 1936.66]; $d_{unb} = 1.11$ [0.45, 1.84]) and post-10' (α-amylase: mean difference with 95% CI = 710.13 [422.34, 997.92]; $d_{unb} = 1.42$ [0.72, 2.24]; protein: mean difference with 95% CI = 1267.61 [611.42, 1923.80]; $d_{unb} = 1.08$ [0.45, 1.79]). Furthermore, a high linear positive correlation was observed between protein concentration and α-amylase activity (mmol/min/L saliva) ($r = 0.9$; $p = 0.03$; $y = 0.4066x + 144.37$). Accordingly, α-amylase activity expressed by mg of protein did not present differences between saliva collections for each individual.

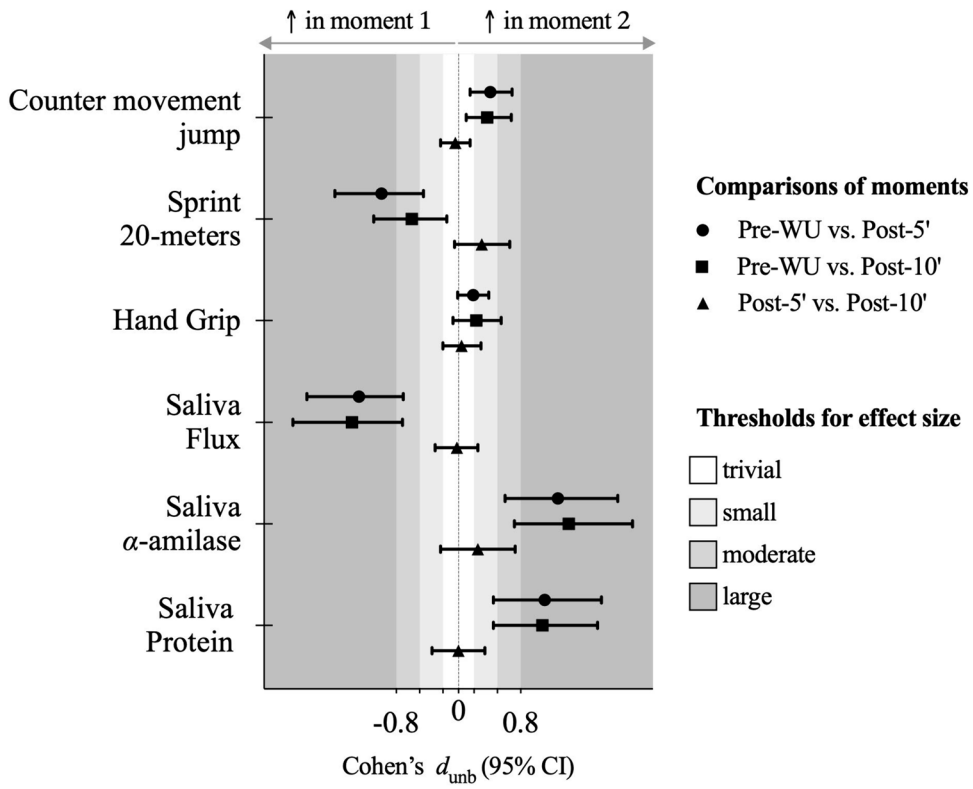

Fig. 4 *Cohen's $d_{unbiased}$* according to time windows comparisons. Error bars indicate uncertainty in true mean changes with 95% confidence intervals. Note: higher time in 20 m sprint protocols is related to worse performance

Discussion

This investigation shows that general team sports' WU may enhance physical performance up to 10 min after its conclusion. Even though vertical jump and sprint may be potentiated during such time window, sprint capacity showed larger improvements at both 5–10 min time points. Interestingly, vertical jump capacity seems to be maintained across 10 min, but sprint ability may decrease over time. These results represent key insights into WU's design, organization and structure.

Vertical jump and sprint capacity improved after WU, although greater effects were observed for sprint. A previous systematic review with meta-analysis has shown that conditioning activities may produce small potentiation effects for jump and moderate effects for sprint performance [8]. Moderate-intensity active WUs may improve short-term jumping and running capacities, a phenomenon strongly linked to increased muscle temperature, metabolism and fiber conduction velocity [25]. In the present WU routine, α-amylase does not appear to have been overexpressed in relation to the total proteins, which can be indicative that the anaerobic threshold was not largely exceeded during WU [16]. Moreover, as for low- to moderate-intensity running, dynamic stretching, bodyweight strength exercises and repeated sprints were included, it was somehow expected to identify improvements in vertical and horizontal oriented activities such as jumping and running. In fact, as jump height is strongly correlated to running velocity [26] and it was not surprising to observe similar trends in both capacities.

Regarding CMJ performance enhancement, the inclusion of vertically oriented exercises such as squats and lunges may have helped promote such improvements. It was already shown that the inclusion of low- to moderate-intensity squats in WU activities promoted acute CMJ performance improvements, partly supported by the greater activity of knee extensor muscles [27]. Adding up, maintaining the movement pattern from conditioning activity to testing (i.e., preservation effects) appears to be important criteria to performance potentiation [5]. However, it should be noticed that the WU protocol used in this study also included dynamic stretching and repeated sprint activities, with and without changes of direction. Even though the motor pattern may be different between these exercises and jumping actions, research shows that dynamic WU, including dynamic flexibility [28] and sub-maximal running activities [29], may also have positive effects on vertical jump performance. Additionally, the inclusion of short-term high-intensity running actions may have contributed to the positive effect on subsequent jumping performance, as it was already shown that low-volume, high-intensity fast stretch–shortening cycle actions involved in WU routines may increase specific fast muscle actions such as jumping [30].

In the present investigation, vertical jump enhancement was maintained during a 10 min time window. Potentiating

time windows after conditioning activities is not consensual because of the multiple factors that may affect the balance between performance enhancement and fatigue [5]. In fact, optimal recovery intervals to maximize subsequent voluntary muscle function may range from 5 to 12 min [31].

Although speculative, another justification for maintaining positive effects on jumping over time may rely on the magnitude of potentiation after WU, which was small and, therefore, easily prolonged in time. Studies reporting moderate acute performance improvements have indicated muscle temperature as the most important mechanism to support potentiation, particularly when short-term force development is key to the performance test [5] as it happens on CMJ. Thus, it is highly possible that participants were able to maintain muscle temperature and muscle contractile properties during the 10 min interval.

Significant improvements were observed in running performance after the WU, mainly at 5 min. In fact, salivary markers revealed an intense sympathetic activation 5 min after WU, with a marked decrease in salivary flux and an increase in total protein concentration, including α-amylase, which is still not reversed 10 min after WU. As previously mentioned, the dynamic nature of the WU is well established in the literature as an important factor to improve short-term running performance. For example, an increase in 20-m sprint performance was already observed after a standard 10 min WU that included low-intensity running and active dynamic stretching in rugby players [32]. In that investigation, the authors suggested core temperature improvement and the representativeness of movement patterns as possible factors to justify the positive performance outcomes. The WU structure applied in our study included an initial part of low-intensity running exercises, followed by dynamic flexibility activities and ended with short-term sprinting actions. This sequence may have played a major role in running acute potentiation, since active WUs, including initial aerobic activities followed by low-volume high-intensity efforts, such as activation sprints, are recommended to enhance and sustain subsequent sprint performances [3, 33]. Moreover, it was already observed that power production and sprinting may increase for up to 10 min after WUs including low-intensity running combined with dynamic exercises such as skipping, walking lunges and high knees [34], which also represent exercise categories included in the present investigation.

Even though sprinting capacity was improved 10 min after the baseline, the highest potentiating point was found at 5 min, which means that potentiating effect may have been attenuated over time. It is known that muscle power production can be highly improved after a preload stimulus when adequate recovery intervals are provided between the conditioning stimulus and the explosive performance activity [35]. Since repeated high-intensity sprints, including accelerations and decelerations activities, were used at the end of the WU, increased muscle temperature and the representativeness of movement patterns may have been the most critical factors contributing to enhanced sprint capacity after 5 min. In fact, traditional moderate- to high-intensity conditioning activities may elicit higher potentiating effects 5 min after the initial stimulus [8]. Also, 10–20 m sprint velocities were already reported to reach their highest potentiation moment 4 min after a plyometric conditioning activity [36]. However, sprint is a highly technical motor skill [37] that requires coordinated arm–leg movements to achieve high acceleration and maximum speed [38]. It is possible that these coordination requirements combined with participants' limited execution technique and lower muscle temperature resultant from a 5 min passive interval (i.e., from 5 to 10 min after WU) may have contributed to attenuated potentiating effects. In summary, this suggests that Re-WU strategies could be employed following the initial WU to maintain physical performance and mitigate potential acute impairments, especially in sprinting. From this perspective, if the transition periods between the conclusion of the WU and the beginning of the match are prolonged, performing short-term explosive tasks could be an effective method to reduce performance impairment and, in some cases, even enhance performance [39]. For instance, in football plyometric exercises, repeated changes of direction may be simple, quick and effective activities to mitigate the decline in sprint capacity after WU [40].

A substantial body of research consistently demonstrates that inter-individual variability is pronounced following acute potentiation protocols, which may be attributed to the fact that athletes possess diverse physical profiles and training backgrounds [10]. The findings of our study also underscore the necessity of recognizing inter-individual differences in all variables, as illustrated in Figs. 2 and 3. It is noteworthy that even with an identical WU and the same time window, unique inter-individual responses can still emerge. This variation suggests that a 'one-size-fits-all' strategy may not be the most effective method for structuring WU. Instead, a personalized approach, informed by meticulous observation and understanding of inter-individual differences, could serve as a valuable addition to coaching strategies. Coaches should be aware that the time necessary to induce the greatest potentiating effects is influenced by the type of the conditioning stimulus, load and training status [5], which means that different responses may emerge over time when the same WU routine is applied to everyone.

The authors acknowledge that the absence of a control group is a limitation for the interpretation of the results of the current study. This issue primarily arose due to logistical constraints and the athletes' availability. Despite this, we strongly believe that the findings of this investigation carry significant practical implications for real-world scenarios.

When compared to vertical jump, sprint capacity may be enhanced at greater extent during a 10 min time window after WU; however, small decrements may be observed over time, which reinforces the importance of considering Re-WU activities when longer passive recovery intervals exist.

Conclusion

Standard WU routines in team sports may acutely enhance jumping and running performances, with the most pronounced effects observed for sprinting at the 5 min time point. While improvements in jumping may persist after 10 min, sprinting ability may experience a slight decline within this time window. These findings highlight the importance of understanding the optimal Post-WU time window in team sports. This knowledge can help coaches to adopt efficient strategies, facilitating a more precise synchronization of the WU conclusion with the start of a match and/or the potential integration of Re-WU strategies when necessary. Moreover, acknowledging the variability in athletes' responses to the same post-WU time window can assist coaches in customizing WU strategies to meet individual needs. Coaches should also be aware that potentiation time windows are influenced by the type of conditioning activity, load and training status.

Author contribution All authors contributed equally.

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Data availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

Ethical approval The investigation was approved by the University of Évora Ethics Committee.

Consent to participate and to publish Written informed consent was obtained from all participants.

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